

4(a)(i)	glycosidic ;	1
4(a)(ii)	any three from: label one monomer correctly ; α-glucose with a hydroxyl group on C1 ; fructose drawn with a hydroxyl group on C2 ; water used in the reaction ;	3
4(b)(i)	sieve plate ;	1

Question	Answer	Marks
4(b)(ii)	any four from: 1 carbohydrate is transported in (phloem) sieve tubes ; A sucrose transported A sieve tube elements 2 as the percentage of outer stem tissue removed increases, the quantity of phloem (sieve tubes) remaining decreases ; 3 less, mass flow / translocation ; 4 0 mg at 100% removal because all phloem removed / AW ; 5 carbohydrate is not transported in xylem tissue, as there is no transport when xylem tissue is left intact ; 6 <i>ref. to</i> movement to lower part of stem from source to sink ; 7 AVP ; e.g. <i>suggestion that</i> most of the (active) phloem is in the inner part of the outer stem <i>idea that</i> phloem sieve tubes are arranged in layers	4

Question	Answer	Marks
4(c)	any four from: max 3 1 example of structural change to sucrase ; e.g. translation stops prematurely / truncated polypeptide / AW change in the primary structure / different amino acid(s) polypeptide does not, fold into / form, (correct) tertiary structure 2 suggestion of why sucrase is non-functional ; e.g. active site, not complementary / changed shape binding site changed shape, sucrose does not bind to active site catalytic site changed shape, glycosidic bond cannot be broken 3 polypeptide not produced as mRNA does not attach to ribosome ; 4 polypeptide recognised as abnormal and degraded ; max 3 5 deletion mutation is loss of one or more nucleotides (in the gene coding for sucrase) ; 6 (causes a) change in the sequence of, nucleotides / bases / base pairs, (in the, gene coding for sucrase / DNA molecule) ; 7 (causes a) change in the sequence of, nucleotides / bases, in, mRNA / an mRNA molecule ; 8 consequence to codons ; (if not a deletion of, three / multiples of three) all the codons after the deletion mutation are altered causes a frameshift (deletion, of three / multiples of three) one or more codons absent, then same sequence 9 mutation may, form a premature stop codon / introduce a stop codon ;	4